

# Google DeepMind Lyria 3: Music Generation Model Outputs 48kHz Stereo with Real-Time Steering and SynthID Watermarking

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The graphic features a dark blue background with a subtle grid. At the top left, the text 'Google DeepMind Lyria 3' is displayed in white and green. Below it, a subtitle reads 'AI music generation with real-time steering and SynthID watermarking'. Three key features are highlighted in separate boxes: '48kHz' (Studio-quality stereo, Full arrangements from scratch), '<2s' (Real-time latency, Steer music as it plays), and 'SynthID' (Inaudible watermark, Survives compression). At the bottom left, a line of text states 'Text, image, or audio → full multi-instrumental tracks with vocals'. A small logo is visible in the top right corner, and the 'ToKnow.ai' watermark is in the bottom right corner.

**Google DeepMind Lyria 3**

AI music generation with real-time steering and SynthID watermarking

- 48kHz**  
Studio-quality stereo  
Full arrangements from scratch
- <2s**  
Real-time latency  
Steer music as it plays
- SynthID**  
Inaudible watermark  
Survives compression

Text, image, or audio → full multi-instrumental tracks with vocals

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Figure 1: Google DeepMind Lyria 3

Google DeepMind released Lyria 3, its most advanced music generation model. The model generates 30-second tracks in studio-quality stereo (48kHz) from text, image, or audio inputs. It does not piece together loops. It generates full musical arrangements from scratch, including vocals and multi-instrumental tracks. Lyria 3 is now integrated into the Gemini app. Describe a mood, genre, or set of instruments, and it outputs a finished audio file. Music generation is hard because it is continuous and multi-layered: melody, harmony, rhythm, and timbre all need to stay coherent from the first second to the last.

The real shift is the Lyria RealTime API. Traditional models work like a jukebox: input a prompt, wait for a file. Lyria RealTime generates audio in 2-second chunks as a live stream. Users can steer style, tempo, and instrumentation while the track plays, with under 2 seconds of delay. The Music AI Sandbox builds on this, transforming a hum into a full orchestral arrangement or turning chord progressions into a vocal choir. Every track includes a SynthID watermark embedded in the audio waveform, inaudible to humans but detectable by software even after MP3 compression or analog re-recording. As AI-generated music floods streaming platforms, permanent attribution solves a real trust problem for creators and listeners. Lyria 3 competes directly with Suno and Udio but brings multimodal input and Google's infrastructure advantage.

Sources:

- [Google DeepMind Lyria Technical Details](#)
- [MarkTechPost: Google DeepMind Lyria 3](#)

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